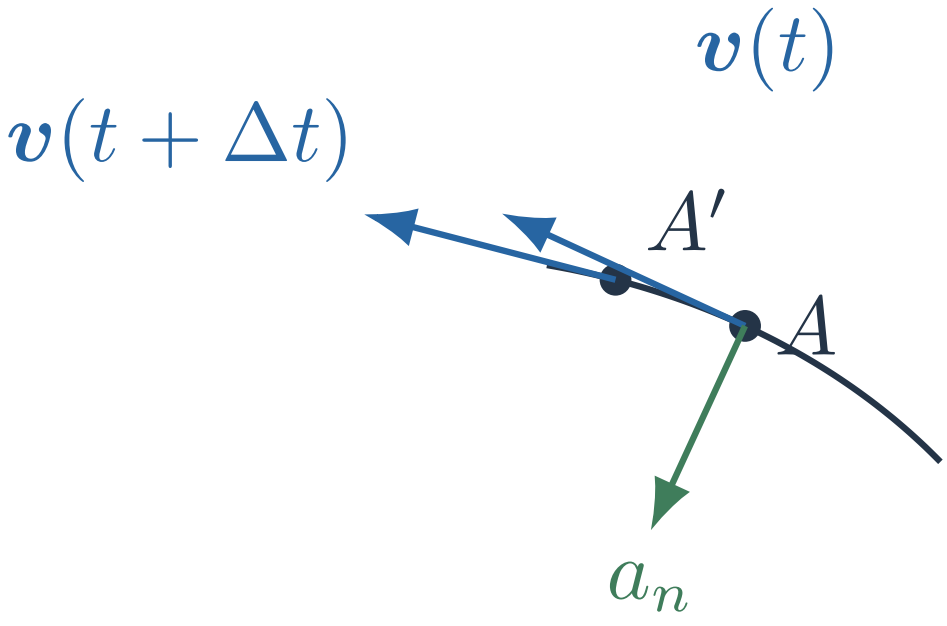

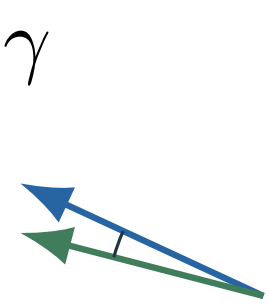

$$\Delta\gamma = \frac{v\Delta t}{l}\cos^3\beta$$

相邻切线转角的示意；计算取 $\Delta t \to 0$